IMO 2026 — Problem 2

Solution by GPT-5.6 Sol (lightweight, xhigh reasoning)

2026-07-18

Source: `results/imo-2026/gpt-5.6-sol-lightweight-xhigh/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle LBA, L lies inside the angle ACK, and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Solution document

Status

partial

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Approaches tried

- **Angle coordinates:** Put $x = \angle KBA = \angle ACL$, $y = \angle LBK = \angle LNC$, and $z = \angle LCK = \angle BMK$. The two small triangles BMK and CNL give explicit coordinates for K and the relation between L and N.
- **Raw Cartesian and complex elimination:** Both verify the theorem, but unorganized expansion produces expressions too long for a rigorous olympiad write-up.
- **Tangent/homogeneous vector factorization:** This gives a compact exact relation between the desired circle determinant and the remaining line-incidence equation.
- **Complex unit-exponential factorization:** This independently gives the same relation. Its one-line "substitute and simplify" proof was too terse.

- **Invariant reorganization (current):** Instead of expanding the coordinates of N, write $L = \lambda N$, reduce both determinants to $K \cdot L$, $[K, L]$, and $|L|^2$, and then compare the four coefficients of the ray parameter. This has reduced the open algebra to a small coefficient table.

Current best

The complete geometric reduction is written below. The factorization lemma is now reduced, for generic y, z, to tangent variables $p = \tan x, q = \tan y, s = \tan z$ and a scaled ray parameter r. Writing $\lambda = U + iV$, $W = |\lambda|^2$, $P = K \cdot L$, $C = [K, L]$, and using the non-unit rotation $T = (1 - ps) + i(p + s)$, direct invariant calculations give

$$WD = C((1-U)|L|^2 - W) + |L|^2\{V(P - |K|^2) - WK_y\} + WL_y|K|^2,$$

$$WQ = C((1-ps)U + (p+s)V) + P((p+s)U - (1-ps)V) - 2(p+s)|L|^2.$$

Here P, C are linear and $|L|^2$ is quadratic in r. It remains only to display and verify the resulting four coefficient equalities proving

$$(p+q)(p+s)D = p(p(1+q^2)r + q)Q.$$

Continuity then handles $\cos y \cos z = 0$ and conversion back gives the desired trigonometric coefficient.

Full proof

We identify the Euclidean plane with $\mathbb{C}$. After a similarity and, if necessary, a reflection, take

$$A = 0, \qquad B = 2, \qquad M = 1,$$

with C in the upper half-plane. Lower-case letters will also denote the complex coordinates of the corresponding upper-case points. Put

$$x = \angle KBA = \angle ACL, \qquad y = \angle LBK = \angle LNC, \qquad z = \angle LCK = \angle BMK.$$

All three numbers are positive. Moreover, the nondegenerate triangles BMK and CNL give

$$0 < x + z < \pi, \qquad 0 < x + y < \pi. \qquad (1)$$

In particular, the sines occurring below are nonzero.

Because K is inside triangle BMC, the ray BK lies above BA. Since K is inside $\angle LBA$ and $\angle LBK = y$, the ray BL is obtained by turning through $x + y$ from BA toward the interior of $\angle ABC$. Hence, with $t = BL > 0$,

$$l = 2 - te^{-i(x+y)}. \qquad (2)$$

In triangle BMK, we have $BM = 1$, the angles at B, M are x, z, respectively, and the angle at K is $\pi - x - z$. The sine rule and the direction of MK therefore give

$$k = 1 + \frac{\sin x}{\sin(x+z)} e^{iz}. \qquad (3)$$

Also $C = 2N$. In triangle CNL, the angles at C, N are x, y: indeed, C, N, A are collinear and $\angle NCL = \angle ACL = x$, while $\angle LNC = y$. Thus the sine rule gives

$$\frac{NL}{CN} = \frac{\sin x}{\sin(x+y)}.$$

The ray NL is obtained from NC by a clockwise turn through y, so

$$l = n + \frac{\sin x}{\sin(x+y)} e^{-iy} n = \lambda n, \qquad \lambda = 1 + \frac{\sin x}{\sin(x+y)} e^{-iy}. \qquad (4)$$

Finally, L lies inside $\angle ACK$ and $\angle ACL = x$, $\angle LCK = z$. Consequently the direction of CK is obtained from that of $CA = -n$ by turning through $x + z$ toward the interior of $\angle ACB$. It follows that $k - 2n$ is a real multiple of $e^{i(x+z)}n$. If

$$[u, v] = \operatorname{Im}(\bar{u}v),$$

this is precisely

$$Q := [k - 2n, e^{i(x+z)}n] = 0. \qquad (5)$$

We shall use the following algebraic identity, whose coefficient verification is being completed below.

Factorization lemma. Under (2)–(4),

$$\begin{aligned} D &:= [k, l](|n|^2 - 1) - [k, n-1]|l|^2 + [l, n-1]|k|^2 \\ &= \frac{\sin x\,(t \sin x + \sin y)}{\sin(x+y)\sin(x+z)} [k - 2n, e^{i(x+z)}n]. \qquad (6) \end{aligned}$$

Assuming the lemma, (5) and (6) yield $D = 0$. It remains to explain why this is the desired conclusion. Let o be the coordinate of the circumcenter O. Since the circle (AKL) passes through the origin, its power function is

$$\mathcal{P}(w) = |w - o|^2 - |o|^2 = |w|^2 - 2\operatorname{Re}(\bar{o}w).$$

The equations $\mathcal{P}(k) = \mathcal{P}(l) = 0$ are

$$2O \cdot K = |k|^2, \qquad 2O \cdot L = |l|^2. \qquad (7)$$

The equality $\mathcal{P}(n) = \mathcal{P}(1)$ is equivalently

$$2O \cdot (N - M) = |n|^2 - 1. \qquad (8)$$

The determinant of the three equations (7)–(8), viewed as equations in the two coordinates of $2O$, is

$$\det \begin{pmatrix} \operatorname{Re} k & \operatorname{Im} k & |k|^2 \\ \operatorname{Re} l & \operatorname{Im} l & |l|^2 \\ \operatorname{Re}(n-1) & \operatorname{Im}(n-1) & |n|^2 - 1 \end{pmatrix} = D.$$

Because A, K, L are noncollinear, the first two rows determine O uniquely; hence $D = 0$ implies (8). Therefore

$$|ON|^2 - |OM|^2 = \mathcal{P}(n) - \mathcal{P}(1) = 0,$$

and so $OM = ON$.

Verification of the factorization lemma

For the moment suppose $\cos y \cos z \neq 0$. Since x is smaller than both $\angle ABC$ and $\angle ACB$, and these two angles cannot both be at least $\pi/2$, we have $0 < x < \pi/2$. Put

$$p = \tan x, \qquad q = \tan y, \qquad s = \tan z, \qquad r = t \cos x \cos y.$$

Then (2)–(4), in Cartesian coordinates, become

$$K = \left(\frac{2p+s}{p+s}, \frac{ps}{p+s}\right), \quad L = (2 - r(1-pq), r(p+q)), \quad L = \lambda N,$$

where

$$\lambda = U + iV, \qquad U = \frac{2p+q}{p+q}, \qquad V = -\frac{pq}{p+q}.$$

Set $W = U^2 + V^2$, $P = K \cdot L$, and $C = [K, L]$. Since $N = \bar{\lambda} L / W$, we have

$$K \cdot N = \frac{UP + VC}{W}, \qquad [K, N] = \frac{UC - VP}{W}, \qquad |N|^2 = \frac{|L|^2}{W}. \qquad (9)$$

Also $[L, N] = -V|N|^2$. Substitution of (9) into the definition of D gives

$$WD = C((1-U)|L|^2 - W) + |L|^2\{V(P - |K|^2) - WK_y\} + WL_y|K|^2. \qquad (10)$$

The non-unit complex multiplier

$$T = (1 - ps) + i(p+s) = \frac{e^{i(x+z)}}{\cos x \cos z}$$

will be used in place of the rotation $e^{i(x+z)}$. If $Q_T = [K - 2N, TN]$, (9) similarly gives

$$WQ_T = C((1-ps)U + (p+s)V) + P((p+s)U - (1-ps)V) - 2(p+s)|L|^2. \qquad (11)$$

The remaining coefficient comparison in (10)–(11) is to be inserted here.